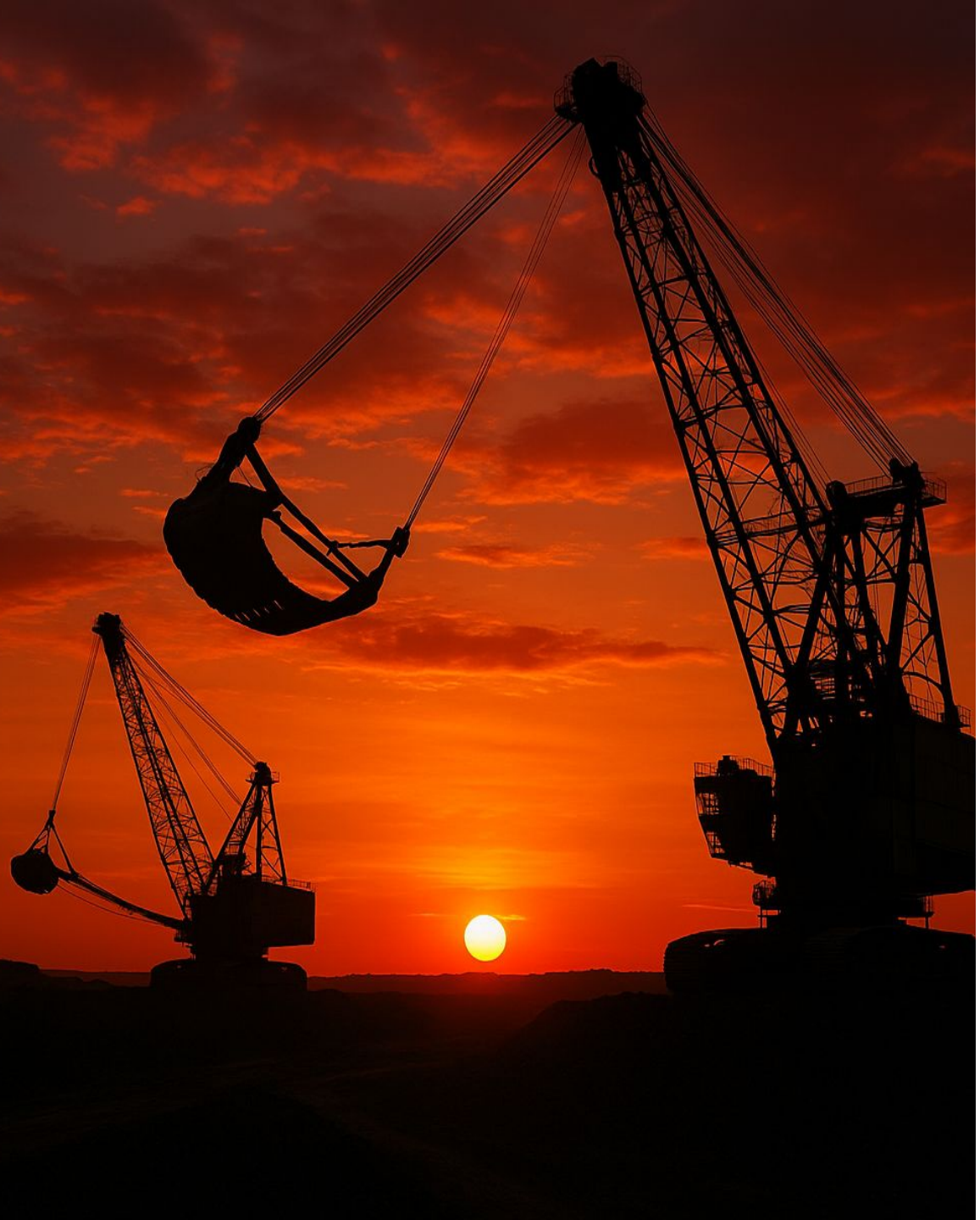
An AI Enabled Platform for Diagnosis, Prognosis and Health Management of Mining Electrical Systems (A Case Study on the Drag Line)



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Sunita Mishra, Assistant Professor, Mining Engineering (Joint Principal Investigator)
Mamata Jenamani, Professor, Industrial and System Engineering (Co-Investigator)
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Objective

To implement a proactive maintenance strategy by leveraging AI and ML-based fault diagnostics and prognostics systems, enabling early fault detection, failure prediction, and maintenance recommendations for a typical dragline.

Modelling	Conducting a comprehensive study of the system and creating a detailed model for the system on the basis of electromagnetics, thermal, strain-field and statistical models to generate data.
Hardware Design	Design and implementation of the sensors and the communication network and deploying a signal processing strategy to gather and analyze data from the system.
Software Design	Developing AI and machine learning algorithms for event detection, diagnosis, prognosis and RUL prediction of faults.

Methodology

Level 3 : Cloud Analytics

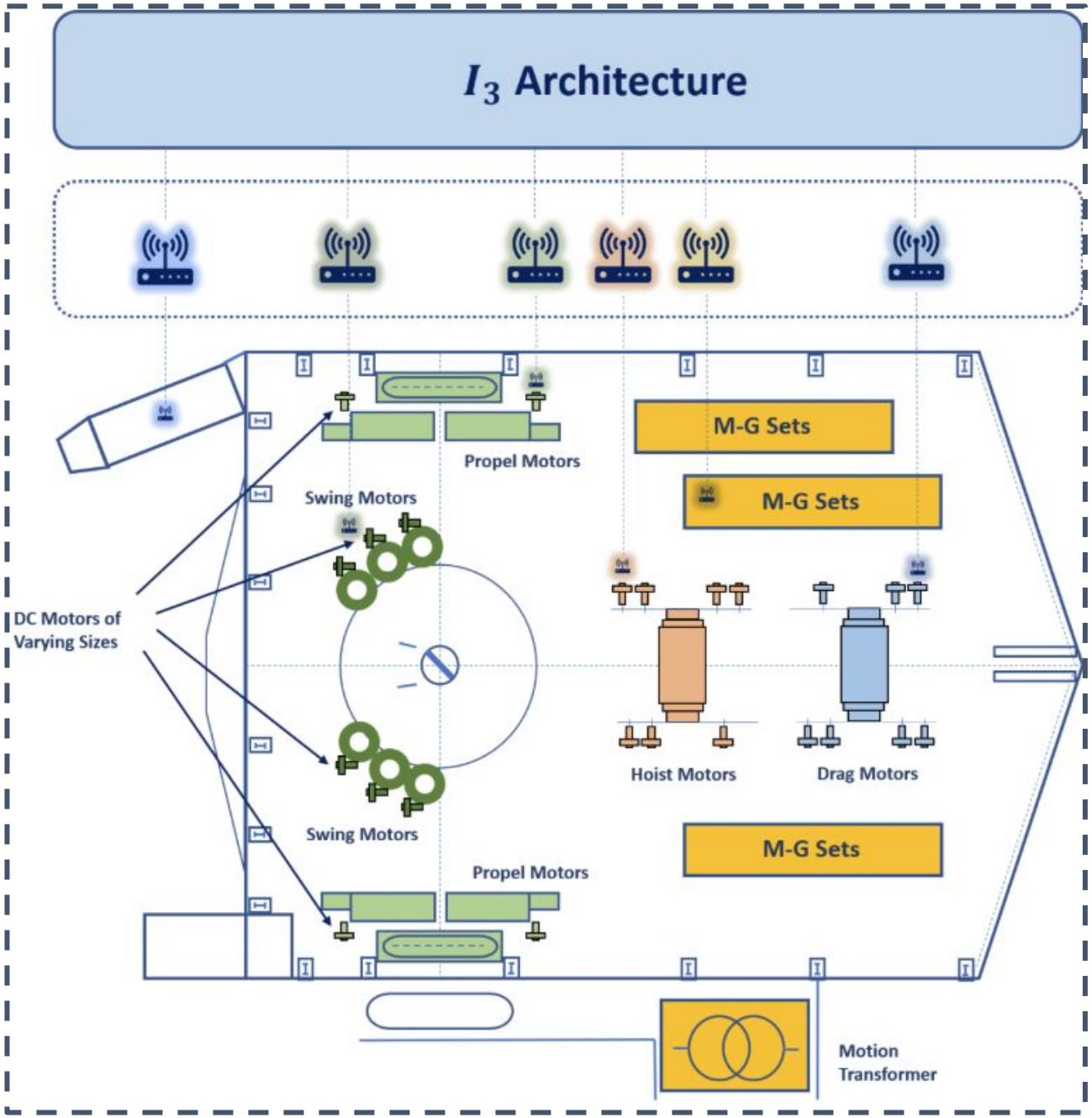
- Analyzes data across systems to predict faults and RUL.
- Generates health cards and schedules maintenance.

Level 2 : Local Aggregation

- Collects data from multiple subsystems.
- Stores locally and provides user interface.

Level 1 : Edge Sensing & Detection

- Captures real-time sensor data and detects anomalies.
- Sends alerts/events to the next level.



Architecture for prognostics and diagnostics of Dragline

Work-Plan

AI-Driven Predictive Maintenance Framework for Dragline System

WE-1: Physics based system modelling and validation (Qtr1,2,3)
- *Electromagnetic Model*
- *Thermal Model*
- *Dynamic Strain Field Model*
- *Statistical and Ensemble models*

WE-2: Establishing the prior from opinionated data and literature survey (Qtr1, Qtr2)
- *On-site Data collection*
- *Field visit and model identification*

WE-3: Design of sensor nodes with edge computing (Qtr1 to Qtr6)
- *Sensor Selection, interfacing and communication*
- *signal processing for detecting and classifying the faults*

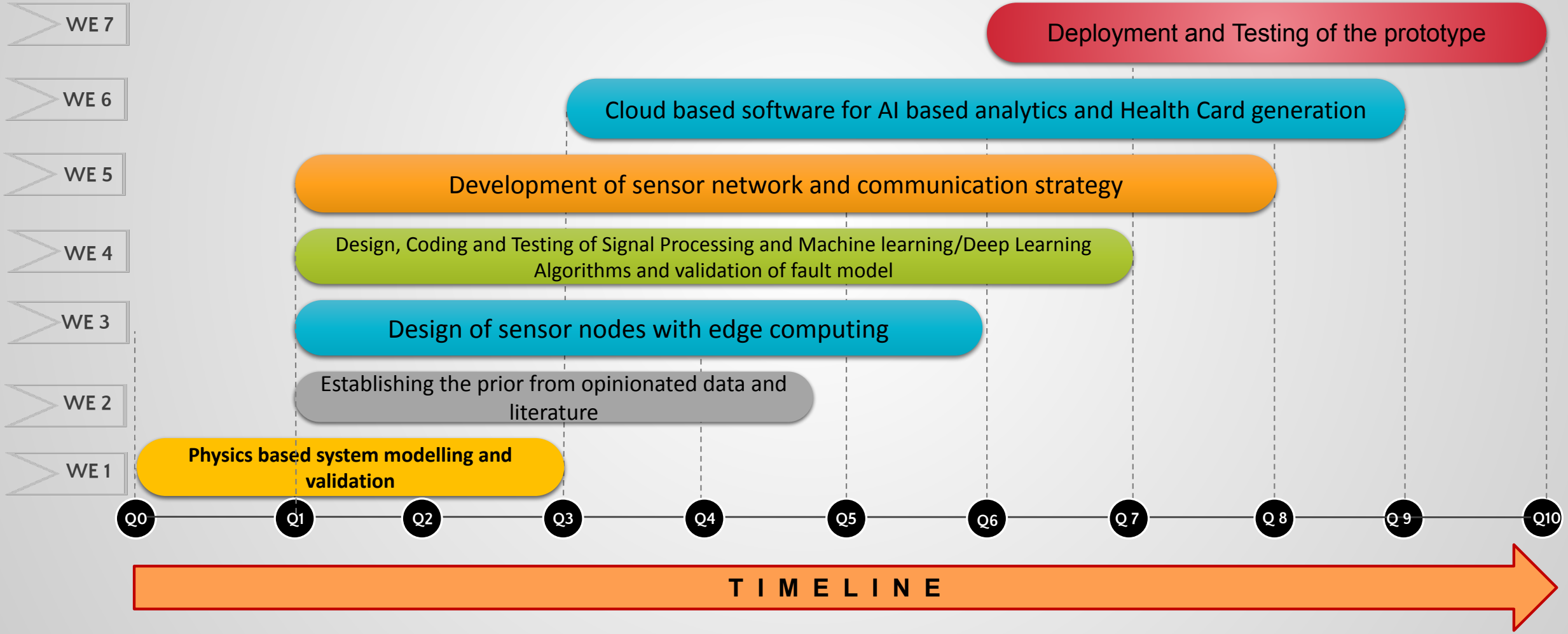
WE-4: Design, Coding and Testing of Signal Processing and Machine learning/Deep Learning Algorithms and validation of fault model. (Qtr1 - Qtr7)

WE-5: Development of sensor network and communication strategy (Qtr1-Qtr8)
Communication strategy and data transfer with Cell phone apps

WE-6: Cloud based software for AI based analytics and Health Card generation (Qtr3-Qtr9)
Development of degradation model

WE-7: Deployment and Testing of the prototype. (Qtr6- Qtr10)
- *Design of prototype*
- *Field and Lab testing*

Timeline


WE 7
WE 6
WE 5
WE 4
WE 3
WE 2
WE 1
Deployment and Testing of the prototype
Cloud based software for AI based analytics and Health Card generation
Development of sensor network and communication strategy
Design, Coding and Testing of Signal Processing and Machine learning/Deep Learning Algorithms and validation of fault model
Design of sensor nodes with edge computing
Establishing the prior from opinionated data and literature
Physics based system modelling and validation
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q 7
Q 8
Q 9
Q10
T I M E L I N E

Work Elements: Overall Progress Summary

Work Element	Planned Timeline	Status	Progress	Lead
WE-1: Physics-based System Modeling	Q1-Q3	In Progress	30%	Prof. SM Prof. SD
WE-2: Domain Knowledge & Literature	Q1-Q2	Ongoing	40%	Prof. MJ
WE-3: Sensor Node Design	Q1-Q6	In Progress	25%	Prof. AR
WE-4: Signal Processing & ML Algorithms	Q1-Q7	Initiated	15%	Prof. MJ
WE-5: Sensor Network Development	Q1-Q8	Planning	10%	Prof. AR

Note: Q1 and Q2 activities are currently ongoing. Multiple work elements are progressing in parallel as per project plan.

WE-1: Physics-based System Modeling

Timeline: Q1-Q3 | Team Lead: Prof. Sunita Mishra

Objective

Develop a comprehensive physics-based model of the dragline electrical system (mechanical, electromechanical, and control subsystems) to generate synthetic data, understand fault signatures, and validate against real-world observations.

Detailed Tasks

- **Task 1.1 — Study of System Operations**
  - Review dragline electrical system documentation and OEM manuals
  - Map all critical electrical subsystems: DC Motors (swing, propel, hoist, drag), M-G Sets, Motion Transformer
  - Identify operating regimes, load profiles, and duty cycles
  - Responsibility: SM (lead), RS-1, RS-2, TWF-1

WE-1: Physics-based System Modeling

Timeline: Q1-Q3 | Team Lead: Prof. Sunita Mishra

- **Task 1.2 — Field Visit and Meeting Domain Knowledge Experts (DKE)**
  - Conduct site visits to Northern Coal Fields Ltd, Singrauli
  - Gather firsthand knowledge from operators and maintenance engineers
  - Document observed fault types, frequencies, and operational challenges
  - Responsibility: SM, AR, RS-1, RS-2, IN-1
- **Task 1.3 — Historical Data Analysis**
  - Collect and analyze historical maintenance records and failure logs
  - Identify common fault patterns, MTBF statistics, and critical failure modes
  - Responsibility: SM (lead), RS-1, RS-2
- **Task 1.4 — Electromagnetic Model Development**
  - Develop analytical electromagnetic models for DC motors (varying sizes)
  - Model armature faults, field winding faults, commutator anomalies
  - Simulate fault signatures in current, voltage, and power spectra
  - Responsibility: AR, RS-1

WE-1: Physics-based System Modeling

Timeline: Q1-Q3 | Team Lead: Prof. Sunita Mishra

- **Task 1.5 — Thermal Model Development**
  - Build thermal equivalent circuit models for all motor types
  - Model heat dissipation under normal and faulty conditions
  - Simulate temperature signatures during insulation degradation and overload
  - Responsibility: SD, RS-2
- **Task 1.6 — Dynamic Strain Field Model**
  - Model mechanical stress and vibration signatures under various fault conditions
  - Link bearing faults, shaft misalignment, and eccentricity to vibration patterns
  - Responsibility: SD (lead), RS-2
- **Task 1.7 — Statistical and Ensemble Models**
  - Develop statistical fault models based on historical data distributions
  - Build ensemble models combining electromagnetic, thermal, and strain models
  - Validate models using cross-validation with historical data
  - Responsibility: AR, MJ, RS-1, RS-2

WE-1: Physics-based System Modeling

Timeline: Q1-Q3 | Team Lead: Prof. Sunita Mishra

- **Task 1.8 — Failure Models of Critical Subsystems**
  - Develop fault taxonomy for swing motors, propel motors, hoist motors, drag motors
  - Model M-G set degradation and motion transformer fault signatures
  - Responsibility: SM (lead), AR, RS-1
- **Task 1.9 — Steady State & Transient Modelling and Validation**
  - Validate all models under steady-state operating conditions
  - Validate models during startup, load transients, and fault events
  - Compare model outputs with field data collected during site visits
  - Responsibility: SM, AR, SD, RS-1, RS-2
- **Task 1.10 — Deep Learning-Based Models**
  - Train DL surrogate models to approximate physics model outputs
  - Use physics-informed neural networks (PINNs) for data augmentation
  - Responsibility: AR, RS-1, RS-2

WE-2: Literature Survey, Prior Knowledge & Opinionated Data

Months 3–9

Timeline: Q1-Q2 | Team Lead: Prof. Mamata Jenamani

Objective

Establish informative priors from domain experts, operators, and comprehensive literature to build knowledge graphs and sentiment-analyzed opinionated data for Bayesian and ML models.

Detailed Tasks

- **Task 2.1 — Comprehensive Literature Review on Dragline Faults & Diagnosis**
  - Systematic review of fault detection methods for DC motors, M-G sets, and drives
  - Review of ESA, MCSA, vibration analysis, thermal monitoring methods
  - Survey of ML/DL methods: CNN, RNN, LSTM, Transformer, GNN approaches
  - Review of digital twin and physics-informed approaches for mining equipment
  - Target: 40+ papers with color-coded categorization across signal processing, ML, DL, digital twin
  - Responsibility: MJ (lead), RS-1, RS-2, RS-3

WE-2: Literature Survey, Prior Knowledge & Opinionated Data

Months 3–9

Timeline: Q1-Q2 | Team Lead: Prof. Mamata Jenamani

- **Task 2.2 — On-site Data Collection and Field Visits**
  - Multiple site visits to NCL Singrauli for primary data collection
  - Interview operators, maintenance engineers, and domain experts (DKE rounds)
  - Collect unstructured observations, anecdotal fault reports, and tacit knowledge
  - Responsibility: SM, AR, MJ, RS-1, RS-2
- **Task 2.3 — Opinionated Data Collection and Sentiment Analysis**
  - Structure expert opinions into quantifiable priors via structured questionnaires
  - Apply NLP/sentiment analysis methods (Prof. MJ's expertise) to text data from experts
  - Consolidate multiple expert opinions into weighted probability estimates
  - Responsibility: MJ (lead), RS-3
- **Task 2.4 — Knowledge Graph Construction**
  - Design knowledge graphs linking fault types, causes, signatures, and maintenance actions
  - Use graph-based representation (aligned with IIT KGP's GraphSAGE expertise)
  - Incorporate both literature-derived and expert-derived nodes and edges
  - Responsibility: MJ (lead), AR, RS-3

WE-2: Literature Survey, Prior Knowledge & Opinionated Data

Months 3–9

Timeline: Q1-Q2 | Team Lead: Prof. Mamata Jenamani

- **Task 2.5 — Event Detection Method Survey**
  - Survey and categorize event detection and anomaly detection algorithms
  - Identify best-performing methods for transient and steady-state motor conditions
  - Responsibility: MJ, AR, RS-1
- **Task 2.6 — Fault Classification Method Survey**
  - Survey supervised, semi-supervised, and unsupervised fault classification methods
  - Benchmark few-shot and transfer learning methods for data-scarce scenarios
  - Responsibility: MJ, AR, RS-2

WE-3: Design of Sensor Nodes with Edge Computing

Months 9–12

Timeline: Q1-Q6 | Team Lead: Prof. Aurobinda Routray

Objective

Design, build, and validate three prototype sensor nodes based on different embedded processors with edge computing capability, capable of real-time signal acquisition, event detection, and wireless communication.

Detailed Tasks

- **Task 3.1 — Sensor Selection and Interface Design**
  - Select sensors for: current (Hall-effect, Rogowski coils), voltage, vibration (MEMS accelerometers), temperature (IR + contact), power quality
  - Design analog front-end: anti-aliasing filters, signal conditioning, ADC interfacing
  - Define sampling rates for each signal type based on fault signature requirements
  - Responsibility: AR (lead), RS-1, TA-1

WE-3: Design of Sensor Nodes with Edge Computing

Months 9–12

Timeline: Q1-Q6 | Team Lead: Prof. Mamata Jenamani

- **Task 3.2 — Prototype-1: 32-bit ARM Microcontroller Node**
  - Design sensor node based on 32-bit ARM Cortex processor
  - Implement real-time OS (FreeRTOS or bare-metal) for multi-sensor data acquisition
  - Develop Level-1 event detection algorithms within MCU compute budget
  - Integrate WiFi/LoRa communication module
  - Responsibility: AR, RS-1, TA-1
- **Task 3.3 — Prototype-2: Raspberry Pi Node**
  - Design sensor node based on Raspberry Pi (RPi 4 or CM4)
  - Implement Python/C++ data acquisition with real-time constraints
  - Deploy lightweight ML inference: ONNX Runtime, TensorFlow Lite
  - Test edge inference of Level-1 signal processing algorithms
  - Responsibility: AR, RS-2, TA-2 — aligning with existing IIT KGP Raspberry Pi deployment work
- **Task 3.4 — Prototype-3: Intel NUC Node**
  - Design higher-compute sensor node using Intel NUC
  - Enable Level-2 algorithms including CNN and GraphSAGE-based inference
  - Test model quantization and pruning for compute-budget compliance
  - Responsibility: AR, RS-1, RS-2

WE-3: Design of Sensor Nodes with Edge Computing

Months 9–12

Timeline: Q1-Q6 | Team Lead: Prof. Mamata Jenamani

- **Task 3.5 — Event Detection Algorithms (Edge)**
  - Implement lightweight event detection: threshold-based, CUSUM, sliding window FFT
  - Test fault detection latency and accuracy vs compute cost tradeoff
  - Optimize algorithms for ARM and RPi compute constraints
  - Responsibility: AR (lead), RS-1, RS-2 — building on MCSA/ESA expertise
- **Task 3.6 — Signal Processing for Fault Classification at Edge**
  - Implement STFT-based current spectrum analysis on edge node
  - Test wavelet transform feature extraction within edge compute budget
  - Responsibility: AR, RS-1, TA-1
- **Task 3.7 — Communication to Cloud**
  - Design and test uplink protocol: data compression, packetization, error correction
  - Integrate with WE-5 communication strategies (WiFi, SMS, LoRa)
  - Responsibility: AR, RS-2, TA-2
- **Task 3.8 — Performance Evaluation**
  - Evaluate accuracy, latency, power consumption, and reliability of all three prototypes
  - Benchmark against project KPIs defined in WE-2 literature survey
  - Document hardware BOM, schematics, and firmware for all three prototypes
  - Responsibility: AR (lead), RS-1, RS-2, TA-1, TA-2

WE-4: Signal Processing & ML/DL Algorithm Design

Months 12–15

Timeline: Q1-Q7 | Team Lead: Prof. Mamata Jenamani

Objective

Design, code, test, and validate signal processing and machine learning / deep learning algorithms for three-level hierarchical fault detection: event detection at edge (Level 1), fault diagnosis at local server (Level 2), and prognosis/RUL at cloud (Level 3).

Detailed Tasks

- **Task 4.1 — Time Domain Feature Extraction**
  - Extract RMS, kurtosis, crest factor, peak-to-peak, skewness from current/vibration signals
  - Implement sliding window analysis for transient detection
  - Responsibility: AR, RS-1
- **Task 4.2 — Frequency Domain Analysis**
  - Implement MCSA-based fault frequency extraction (rotor bar, bearing, eccentricity)
  - Automate slot harmonic analysis and sidebands detection
  - Build spectral feature vectors for ML-based classifiers
  - Responsibility: AR (lead), RS-1 — building on MATLAB slot harmonic analysis code

WE-4: Signal Processing & ML/DL Algorithm Design

Months 12–15

Timeline: Q1-Q7 | Team Lead: Prof. Mamata Jenamani

- **Task 4.3 — Time-Frequency Analysis**
  - Implement STFT-based spectrogram generation for CNN input
  - Implement wavelet packet decomposition for multi-resolution fault analysis
  - Test on both steady-state and transient operating conditions
  - Responsibility: AR, SD, RS-1, RS-2
- **Task 4.4 — Machine Learning Algorithms for Feature-Based Fault Detection**
  - Implement SVM, Random Forest, XGBoost for tabular feature-based classification
  - Hyperparameter optimization and cross-validation
  - Benchmark classification accuracy vs computational cost
  - Responsibility: MJ (lead), RS-3
- **Task 4.5 — Deep Learning Architecture for Fault Detection and Classification**
  - Implement 1D-CNN for direct time-series fault classification
  - Implement 2D-CNN operating on STFT spectrograms
  - Implement MobileViT-S backbone with prototypical metric learning (ProtoMobileNet)
  - Implement GraphSAGE encoder for graph-structured multi-sensor data
  - Responsibility: AR (lead), RS-1, RS-2

WE-4: Signal Processing & ML/DL Algorithm Design

Months 12–15

Timeline: Q1-Q7 | Team Lead: Prof. Mamata Jenamani

- **Task 4.6 — Few-Shot and Transfer Learning for Data-Scarce Conditions**
  - Implement prototypical networks for few-shot fault classification
  - Test domain adaptation from lab data to field conditions
  - Responsibility: AR, RS-1 — core IIT KGP research contribution
- **Task 4.7 — Level 1 Algorithms (Edge Node)**
  - Lightweight anomaly detection: statistical process control, threshold-based, CUSUM
  - Optimize for latency < 10ms on ARM/RPi hardware
  - Responsibility: AR, RS-2, TA-2
- **Task 4.8 — Level 2 Algorithms (Local Server)**
  - Multi-fault classification using ML and lightweight DL models
  - Multi-sensor data fusion algorithms
  - Local database query and alerting logic
  - Responsibility: MJ (lead), AR, RS-1

WE-4: Signal Processing & ML/DL Algorithm Design

Months 12–15

Timeline: Q1-Q7 | Team Lead: Prof. Mamata Jenamani

- **Task 4.9 — Level 3 Algorithms (Cloud)**
  - Degradation modelling and Remaining Useful Life (RUL) prediction
  - Cross-sectional and longitudinal health trend analysis
  - Maintenance scheduling optimization based on predicted RUL
  - Responsibility: MJ (lead), SD, RS-3
- **Task 4.10 — Testing with Historical Field Data**
  - Validate all Level 1/2/3 algorithms against historical fault data from NCL Singrauli
  - Compute sensitivity, specificity, F1-score, and false alarm rates
  - Responsibility: SM (lead), AR, MJ, RS-1, RS-2

WE-1: Physics-based System Modeling

In Progress | 30%

Timeline: Q1-Q3 | Team Lead: Prof. Sunita Mishra and Sunil Manohar Dash

Work Completed in Q1-Q2

Electromagnetic Model

- ✓ Electromagnetic Motor model is developed from lab setup Specifications
- ✓ Coupled Circuit model for Induction motor is developed
- ✓ Fault signature analysis framework established
- ✓ Current signature analysis methodology defined

Thermal Model

- ✓ Thermal hotspot locations identified
- ✓ Heat generation zones mapped for motor windings
- ✓ Cooling system parameters documented
- ✓ Temperature distribution model framework created
- ◐ Thermal-electric coupling analysis

Strain Field Model

- ✓ Vibration measurement locations identified
- ✓ Mechanical stress point analysis completed
- ✓ Dynamic load characterization initiated
- ◐ Bearing fault vibration patterns under study
- ◐ FEM model development in progress

WE-2: Domain Knowledge & Literature Survey

Ongoing | 40%

Timeline: Q1-Q2 | Team Lead: Prof. Mamata Jenamani

Work Completed in Q1-Q2

Literature Review Accomplishments

✓ **Comprehensive review of 180+ research papers**

Topics covered:

- Motor fault diagnostics (50+ papers)
- Induction Motor failure analysis (35+ papers)
- AI/ML in predictive maintenance (60+ papers)
- IoT sensor networks (35+ papers)

✓ **Event detection methods catalogued**

- Thermal events, electrical anomalies, mechanical vibrations

✓ **Fault classification methodologies reviewed**

- Traditional: FFT, wavelet analysis
- ML-based: SVM, Random Forest, Neural Nets

Field Data & Expert Knowledge

✓ **Historical data obtained from Literature**

- maintenance records
- Failure logs with root cause analysis
- Downtime statistics and patterns
- Critical fault modes and high-impact components identified

◐ **Knowledge graph construction ongoing**

- Fault-symptom relationships being mapped
- Integration of historical data with domain knowledge
- Validation of relationships using expert inputs and literature

WE-3: Sensor Node Design with Edge Computing

In Progress | 25%

Timeline: Q1-Q6 | Team Lead: Prof. Aurobinda Routray

Work Completed in Q1-Q2

Electromagnetic Model

✓ **Vibration Sensors:**
- IEPE accelerometers (100 mV/g sensitivity)
- Frequency range: 0.5 Hz - 10 kHz
- IP67 rated for harsh environment

✓ **Temperature Sensors:**
- RTD Pt100 (Class A accuracy)
- Infrared thermal cameras (FLIR series)
- Range: -40°C to 350°C

✓ **Current/Voltage Sensors:**
- Hall effect current sensors (200A range)
- High-precision voltage dividers

Edge Computing Hardware Design

✓ **Processor Platform Selection:**
- ARM Cortex-M7 based design (primary)
- Raspberry Pi 4 alternative architecture
- Intel NUC for complex processing

✓ **Interface Design:**
- Multi-channel ADC (24-bit, 8 channels)
- Signal conditioning circuits designed
- Anti-aliasing filters specified

◐ **PCB Layout:**
- Schematic design 70% complete
- Component sourcing in progress

Prototype Development Status

✓ Breadboard prototype of ARM-based node completed - Basic data acquisition validated

◐ Testing with simulated motor signals ongoing - Noise filtering algorithms being refined

◐ Communication module integration in progress - WiFi and LoRa options being evaluated

◐ Final PCB fabrication planned for Q3 - After algorithm validation on breadboard

WE-4: Signal Processing & ML Algorithms

Initiated | 15%

Timeline: Q1-Q7 | Team Lead: Prof. Mamata Jenamani

Work Initiated in Q1-Q2

Signal Processing - Edge Level

✓ **Time Domain Analysis:**
- RMS calculation algorithms implemented
- Peak detection methods coded
- Tested on synthetic motor data

✓ **Frequency Domain:**
- FFT algorithms optimized for ARM
- Spectral analysis framework created
- Harmonic detection logic developed

◐ **Time-Frequency Analysis:**
- Wavelet transform being implemented
- STFT for transient analysis initiated

◐ **Anomaly Detection:**
- Threshold-based methods planned
- Statistical process control approach

Machine Learning Algorithms

✓ **Dataset Preparation:**
- Feature engineering framework defined
- Synthetic data generation for rare faults from Physics Based Models

◐ **Classical ML Models:**
- Preliminary results: 82% accuracy

◐ **Deep Learning Architecture:**
- CNN architecture designed for vibration
- LSTM for time-series prediction initiated
- Model training infrastructure setup

◐ **RUL Prediction:**
- Degradation modeling approach defined
- Implementation planned for Q3

◐ Knowledge graph construction ongoing
- Fault-symptom relationships being mapped

WE-5: Sensor Network & Communication Strategy

Planning | 10%

Timeline: Q1-Q8 | Team Lead: Prof. Aurobinda Routray

Initial Planning & Design Work in Q1-Q2

Communication Technologies Evaluated

✓ **WiFi (IEEE 802.11):**
- Feasibility study completed
- Range: 100m indoor, 300m outdoor
- Bandwidth: Sufficient for streaming data
- Concern: Interference in mining environment

✓ **LoRaWAN:**
- Long-range capability (up to 5 km)
- Low power consumption advantage
- Limitation: Lower data rate (50 kbps)
- Best for: Low-frequency updates

◐ **SMS/Cellular:**
- For alerts and critical notifications
- Backup communication channel
- Cost analysis in progress

Network Architecture Design

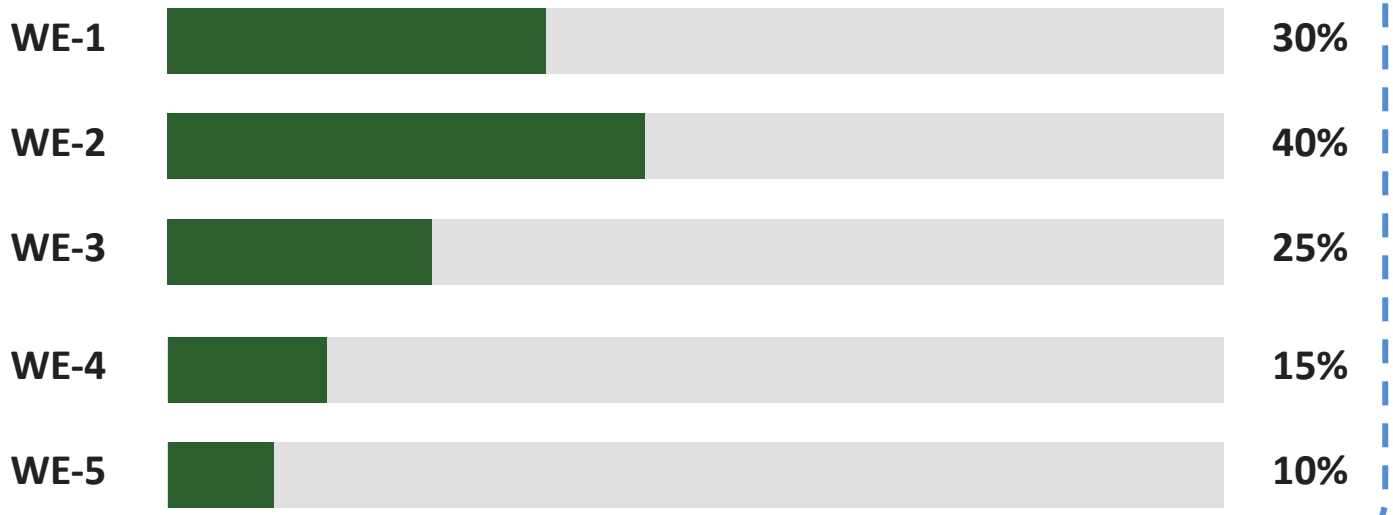
✓ **Hierarchical Architecture Defined:**
- Tier 1: Sensor nodes (edge devices)
- Tier 2: Gateway/aggregator nodes
- Tier 3: Local server at mine site
- Tier 4: Cloud server at IIT Kharagpur

✓ **Data Flow Strategy:**
- Real-time: Critical alerts (WiFi/cellular)
- Periodic: Health metrics (every 5 min)
- Batch: Detailed logs (hourly via LoRa)

◐ **Network Topology:**
- Star topology for WiFi nodes
- Mesh network for LoRa redundancy
- Topology simulation in progress

Work Elements: Overall Progress

Summary



Currently Ongoing Activities (Q2)

- Model validation with historical failure data from Literature
- Integration of electromagnetic-thermal-mechanical models
- Deep learning model development for fault pattern recognition

Prototype Development Status

✓ Breadboard prototype of ARM-based node completed - Basic data acquisition validated

◐ Testing with simulated motor signals ongoing - Noise filtering algorithms being refined

◐ Communication module integration in progress - WiFi and LoRa options being evaluated

◐ Final PCB fabrication planned for Q3 - After algorithm validation on breadboard

Summary: Q1-Q2 Progress

✓ Milestones Achieved (Q1-Q2)

- ✓ **M1:** Project lab setup initiated
- ✓ **M2:** Comprehensive literature review completed
- ✓ **M3:** Physics-based modeling framework established
- ✓ **M4:** Sensor specifications finalized
- ✓ **M5:** Breadboard prototype demonstrated
- ✓ **M6:** Initial ML models trained with baseline accuracy

◐ Upcoming Milestones (Q3)

- ◐ **M8:** Complete electromagnetic model validation
- ◐ **M9:** Finalize PCB design and fabricate first batch
- ◐ **M10:** Deploy prototype sensor network (5 nodes)
- ◐ **M11:** Achieve 85%+ fault classification accuracy
- ◐ **M12:** Establish cloud data pipeline
- ◐ **M13:** Conduct field trial at NCL dragline site
- ◐ **M14:** Generate first automated health report

Summary: Q1-Q2 Progress

- ✓ **All five work elements initiated and progressing as per plan**
  - ✓ WE-2 (Literature & Domain Knowledge) - 40% complete, ahead of schedule
  - ✓ WE-1 (Modeling) - 40% complete with validated electromagnetic framework
  - ✓ WE-3 (Sensors) - 25% complete with finalized specifications

Project positions have been advertised and the recruitment process has been initiated

Next Phase (Q3): Model validation, prototype testing, network deployment

THANK YOU